SOIL EROSION AND CONSERVATION · NOTE ER06

Runoff — Terms, Controlling Factors, Design Rates and Time of Concentration

Course topics: SWCE3.5 L10, SWCE-502 T2, SWCE-502 T5, SWCE7.4 L2

What runoff is

Runoff is the part of precipitation that flows to streams, lakes or oceans as surface or subsurface flow — usually meaning surface flow from rain. Planning conservation and drainage works needs its peak rate, volume and distribution in time.

- Before runoff starts, rain must satisfy evaporation, infiltration, surface storage, surface detention and channel detention. In light rain, interception may keep the soil dry; dense forest or shrub cover can intercept ~25% of annual precipitation.
- Runoff begins when rainfall intensity exceeds infiltration: depressions fill, overflow, and overland flow moves toward streams.
- **Initial detention (depression storage):** water needed to fill depressions before runoff. **Detention storage (surface detention):** water held temporarily on the land while moving to streams.
- **Surface runoff** flows over land to the channels; **subsurface flow** is infiltrated water moving laterally and emerging lower down — common where permeable topsoil overlies an impervious layer.

Term	Meaning
Runoff	Portion or discharge of precipitation appearing in streams
Total runoff	Runoff from all sources over an area for a period or storm
Cumulative runoff	Runoff summed over time
Mean annual runoff	Average annual stream discharge from an area
Depth of runoff	Total runoff ÷ watershed area
Rate of runoff	Volume draining per unit time
Runoff coefficient	Maximum runoff rate ÷ uniform rainfall intensity lasting at least the time of concentration
Percentage runoff	Runoff as % of rainfall
Annual yield	Mean annual runoff volume of the watershed
Runoff cycle	Part of the hydrologic cycle from precipitation to stream discharge or evapotranspiration
Precipitation	All atmospheric moisture (rain, snow, dew, hail…) as depth of water
Point rainfall	Rainfall at a single station
Priming rain	Rain that produces no runoff
Rainfall distribution coefficient	Maximum point rainfall ÷ basin mean rainfall
Storm (rainstorm)	Meteorological disturbance producing precipitation
Rainfall excess (net rainfall)	Rainfall available for runoff
Storm runoff	Runoff reaching the measuring point soon after rain (base flow excluded)
Overland flow	Runoff moving over land before entering streams
Surface runoff	Overland flow after it reaches a definite channel
Surface flow	Stream flow on or above the earth's surface
Subsurface flow (interflow)	Lateral flow below the surface but above the water table
Watershed (drainage basin, catchment)	Area contributing surface flow to a stream, river, reservoir or lake
Drainage coefficient (modulus)	Runoff depth removed from an area in 24 hours

Factors affecting surface runoff

Climatic factors

- Precipitation (intensity, duration, areal distribution, storm pattern), evaporation and evapotranspiration, and wind.
- Higher intensity → more runoff (infiltration capacity unchanged). Longer duration → more runoff as infiltration declines. Runoff peaks when the whole watershed contributes at once.
- A storm near the outlet gives more runoff than the same storm upstream; intense rain late in a storm gives high runoff (infiltration is low by then); even mild, long rain can give considerable runoff; a storm moving down the slope gives a higher peak sooner.

Physiographic factors

- **Watershed:** size, shape and orientation, land use, soil infiltration, topography, vegetation and conservation practices.
- A larger watershed yields more runoff volume but less per unit area. A fan-shaped watershed peaks higher than a long, rectangular one. Flat land and depressions shed less than steep, well-drained slopes. Forest with mulch cuts runoff substantially.
- Soil moisture, humus, organic matter and temperature strongly control runoff: wet soil infiltrates little; after a long dry spell rain first refills the soil; successive monsoon rains raise runoff substantially.
- **Channels:** size, cross-section, shape and roughness of bed and sides; vegetation in channels retards flow, causes detention and lowers the peak; channel length and stream density matter.
- Small upstream watersheds — the focus of soil conservation — respond to soil and vegetation; large ones are dominated by channel storage and stream hydrology.

$$\text{Drainage density} = \text{Total channel length} / \text{Watershed area}$$

Higher drainage density → lower peak runoff, because the flow spreads over many channels (as stated in the notes).

Stream flow combines channel interception of precipitation, overland flow (direct runoff), subsurface flow (interflow) and groundwater flow (base flow); the hydrograph is the watershed's integrated response.

Design runoff rates

The design runoff rate is the capacity a conservation or drainage structure needs for the expected runoff of a chosen return period: about the **10-year** runoff for vegetated waterways and temporary structures, and the **50–100-year** runoff for permanent structures. The choice balances cost against possible downstream damage.

Time of concentration (Tc)

- The time runoff takes to travel from the most remote point of the watershed to the outlet. When a storm lasts Tc, the whole watershed contributes at once and discharge peaks; rain must last at least Tc for full effect.
- Tc grows with watershed size and dense vegetation (which retards flow) and shrinks with steeper slopes.

$$\text{Simple estimate: } Tc = \text{Flow distance} / \text{Average velocity}$$

Channel slope (%)	Average velocity (m/s)
1–2	0.60
2–4	0.90
4–6	1.20
6–10	1.50

Kirpich (1940): $Tc = 0.0195 \cdot L^{0.77} \cdot S^{-0.385}$ Tc in minutes, L = maximum flow length (m), $S = H/L$

Alternative form: $Tc = 0.0195 \cdot K^{0.77}$, $K = \sqrt{(L^3/H)}$

Small watersheds (with overland flow): $Tc = 0.0195 \cdot L^{0.77} \cdot S^{-0.385} + (2 \cdot L_0 \cdot n/S)^{0.467}$

L_0 = overland flow length (m), S = slope (m/m), n = Manning's roughness of the land surface. (The two Kirpich forms are identical: $L^{0.77} \cdot (H/L)^{-0.385} = (L^3/H)^{0.385}$.)

Illustration (not in the notes)
L = 1500 m, fall H = 30 m → S = 0.02
$Tc = 0.0195 \times 1500^{0.77} \times 0.02^{-0.385} = 0.0195 \times 279.0 \times 4.51 \approx 24.5$ min

Land surface	Manning's n
Smooth impervious	0.02
Smooth bare packed soil	0.10
Poor grass / row crops	0.20
Pasture (average grass)	0.40
Timber with undergrowth	0.60
Dense forest litter / grass	0.80

One-hour rainfall is the maximum 1-hour rainfall intensity for a given return period. The rational method needs the intensity for a duration equal to Tc, so convert the 1-hour value with the graphical relation, keeping the same recurrence interval.

Sources: ICAR eCourse 'Soil and Water Conservation Engineering' — Dr A. Mishra, reviewed by Dr B. C. Mal (IIT Kharagpur); AgriMoon e-book. Kirpich, Z. P. (1940), Time of concentration of small agricultural watersheds, Civil Engineering 10(6): 362. Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. Time of concentration is the time for runoff to travel from
(a) the outlet to the divide
(b) the most remote point to the outlet
(c) the centroid to the outlet
(d) rain start to peak

Q2. Kirpich's formula gives Tc in
(a) seconds
(b) minutes
(c) hours
(d) days

Q3. Permanent soil-conservation structures are designed for runoff of return period
(a) 2 yr
(b) 10 yr
(c) 50–100 yr
(d) 1000 yr

Q4. Vegetated waterways are designed for about the
(a) 2-year
(b) 10-year
(c) 50-year
(d) 100-year

Q5. Rain that produces no runoff is called
(a) priming rain
(b) net rain
(c) point rain
(d) excess rain

Q6. Compared with a long rectangular watershed, a fan-shaped one gives

(a) a lower peak
(b) a higher peak
(c) no difference
(d) no runoff

Q7. Average runoff velocity for a channel slope of 4–6% is about

(a) 0.60 m/s
(b) 0.90 m/s
(c) 1.20 m/s
(d) 1.50 m/s

Q8. Manning's n for pasture (average grass) overland flow is

(a) 0.02
(b) 0.10
(c) 0.40
(d) 0.80

Answers

Q1 — (b) the most remote point to the outlet
Q2 — (b) minutes — *With L in metres.*
Q3 — (c) 50–100 yr
Q4 — (b) 10-year — *runoff.*
Q5 — (a) priming rain
Q6 — (b) a higher peak
Q7 — (c) 1.20 m/s
Q8 — (c) 0.40